

Cameron DeBough

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EDUCATION

University of Arizona, B.S. Computer Science & Artificial Intelligence, Minor in Mathematics

Expected May 2029

GPA: 3.87 · Dean's List with Distinction (Fall 2025) · Dean's List (Spring 2026) · 67 Credits Completed

Coursework: Software Development, Discrete Mathematics for CS I-II, Analyzing Data, Web Programming, Calculus I-II, Linear Algebra

EXPERIENCE

Founder & Engineer, DeBough Systems (deboughsystems.com)

August 2026 - Present

- Designed, built, and shipped a live security posture product for businesses with no IT staff: Next.js and TypeScript app with Supabase auth and Postgres, Stripe billing, and automated weekly re-scans
- Engineered a passive outside-in scanning engine covering SPF, DMARC, DKIM, TLS validity, domain registration via RDAP, and registered lookalike domains, with transparent scoring and plain-English remediation steps
- Hardened the app with a strict nonce-based CSP, security headers, SSRF-safe outbound networking, and rate limiting; gated breach monitoring behind DNS TXT ownership proof so it cannot be misused for reconnaissance

Research Programmer, NewFOS Center, University of Arizona

March 2026 - Present

New Frontiers of Sound, an NSF Science & Technology Center

- Built Python experiment-automation pipelines for nonlinear acoustics research (sweep-rate hysteresis test suite, harmonic SNR analysis, and transfer-function extraction) supporting Volterra-kernel characterization of transducer nonlinearity
- Designed and validated an electrical-impedance characterization rig with a full de-embedding pipeline that isolated a 42% cable-parasitic error in a previously reported measurement
- Developed MATLAB SVD and Hankel-DMD signal-analysis pipelines producing publication-quality figures for phi-bit (acoustic qubit analogue) experiments on a three-rod coupled acoustic waveguide
- Automated Tektronix oscilloscopes and function generators via VISA/SCPI for synchronized multi-channel acquisition

Undergraduate Research Assistant, VSI Lab, University of Arizona ECE

September 2025 - Present

Embodied AI & Accessibility · Advisor: Prof. Eungjoo Lee

- Building Gaussian-splatting 3D reconstruction and NVIDIA Isaac Sim environments with ROS 2 control of LeRobot SO-101 arms and TurtleBot4 navigation
- Engineering wearable computer-vision pipelines with Meta Aria 2 smart glasses (hand tracking, gaze data, and real-time ML inference on edge devices) for ASL accessibility and skill-assessment applications

Software Engineering Intern, SkillfullyAware / SAAQ

January 2025 - Present

- Own the production backend of an AI-powered leadership diagnostic platform serving EO/YPO organizations: Python and FastAPI services with Supabase (PostgreSQL), deployed and live in production
- Built a streaming LLM analysis pipeline with a multi-stage scoring model and two-pass generation to eliminate timeouts and truncation, plus automated end-to-end DOCX report generation from NLP analysis results
- Implementing authentication, Stripe payments, and persistent user storage for MVP launch

TECHNICAL PROJECTS

APEX QUANT, Algorithmic Futures Trading System

- Architected event-driven backtesting and paper-trading engine for micro futures (MES/MNQ/MGC) with modular strategy framework, risk management controls, and a broker-integration layer targeting Interactive Brokers (ib_insync)

Thermal-Tracking Turret, Embedded Systems / Robotics

- Built an autonomous heat-tracking pan/tilt turret on Raspberry Pi: AMG8833 8x8 thermal array over I²C, PCA9685 hardware PWM servo control, and a closed-loop feedback algorithm that centers a moving heat source in real time, deployed as a headless systemd service

Synapse Finance, Full-Stack Financial SaaS Platform

- Shipped production app with 50+ active users: Flask, React, PostgreSQL, JWT authentication, RESTful APIs, and CI/CD

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript, C#, MATLAB, SQL, Bash, Golang, HTML/CSS

Backend & Frontend: Flask, FastAPI, Node.js, React, Next.js, Tailwind CSS, RESTful APIs, Microservices, Concurrent Programming

Embedded & Hardware: Raspberry Pi, I²C sensor integration, servo/PWM control (PCA9685), instrument automation (VISA/SCPI), real-time control loops, ROS 2

Data & ML: Signal Processing (FFT, SVD, DMD), Machine Learning, Computer Vision, NLP, NumPy/SciPy, Pandas

DevOps & Databases: Docker, Git/GitHub, CI/CD, Linux, AWS, PostgreSQL, Supabase, MySQL, Firebase, Database Design

LEADERSHIP & ACTIVITIES

Coke Fellow, The Coca-Cola Company, Introships Career Discovery Program

May - July 2026

- Selected as 1 of 50 fellows nationwide for a coached cohort program with Coca-Cola executives

University of Arizona AI Club · MLH Global Hack Week · 13+ GitHub projects · Certiport Java Foundations, JavaScript Specialist